

Lab 2 – CreditTrax Product Specification

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Version 2

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for CreditTrax, a mobile application for financial management and literacy. It serves as a technical reference for developers, outlining system behaviors, data processing logic, and integration points necessary for implementation. This document ensures a clear understanding of how CreditTrax will process financial data, generate personalized budgeting recommendations, and support users in managing debt effectively.

1.2 Scope

The purpose of this system is to provide users with a comprehensive financial management platform focused on debt tracking, budget recommendations, and financial literacy tools (such as the “What-if” Analysis feature). By offering structured guidance and actionable insights, the system aims to help users gain better control over their finances and make informed financial decisions.

The system will enable users to track and manage their debts efficiently by allowing them to add, edit, and monitor outstanding balances. The AI could assist users with budget recommendations based on their input, such as income, expenses, and debts. While it cannot fully generate a budget on its own, it will guide users in creating their own budget, helping them optimize their financial plans.

Additionally, the platform will incorporate financial literacy tools to educate users on responsible money management. Through OpenAI-powered “What-If” analysis, users will gain insights into potential financial outcomes based on different scenarios, allowing them to anticipate future financial situations.

To encourage positive financial habits, the system will feature a rewards mechanism where users earn badges for achieving financial milestones. Furthermore, it will support goal management, allowing users to set, track, and modify both short-term and long-term financial objectives.

While the system offers comprehensive financial tracking and guidance, it will not provide direct financial advising, investment recommendations, or automated debt repayment services. Additionally, it does not guarantee improvements in credit scores or financial outcomes. Instead, the platform serves as an empowering tool that equips users with the necessary resources to navigate their financial journey effectively.

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1.3 Definitions, Acronyms, and Abbreviations

AI: Refers to algorithm-driven decision-making processes that analyze user financial data to generate personalized budgeting recommendations and predictive financial insights. It utilizes rule-based algorithms and pre-trained AI models (such as OpenAI's API) to perform “What-If” Analysis and budget optimization based on income, expenses, and debt obligations.

API: A tool that allows the application to connect with other services to integrate extra features.

Credit Score: A three-digit number that reflects an individual’s creditworthiness based on their credit history, influencing lenders' decisions on loan approvals and interest rates.

Debt: Money that is borrowed and must be repaid, typically with interest.

Financial Literacy: A strong understanding of essential financial skills and concepts, such as budgeting, saving, and debt management.

“What-if” Analysis: A technique that allows users to simulate various (financial) scenarios and visualize their potential outcomes.

UI: Refers to the visual and interactive elements that allow users to navigate the application and engage with its features, e.g. buttons, icons, and graphs.

Young Adults: Individuals between the ages of 18 to 34 who are in the workforce and have limited experience with personal finances.

Young Professionals: Young adults aged 18-34 who are in the workforce, in college, or have recently graduated.

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1.5 Overview

The remaining sections of this Software Requirements Specification provide a detailed breakdown of CreditTrax's design and functionality. Section 2.1 outlines the system's role within its broader environment and includes a Major Functional Component Diagram to illustrate its architecture. Section 2.2 highlights key features, specifying which are fully or partially implemented in the prototype. Section 2.3 defines user roles, expertise levels, and typical usage scenarios. Section 2.4 identifies technical, legal, or environmental limitations impacting the system. Finally, Section 2.5 lists external services, APIs, and frameworks required for development.

2 Overall Description

2.1 Product Perspective

CreditTrax is a financial management and educational application designed to assist young adults and professionals in managing their personal finances effectively. The system integrates multiple components to provide users with a comprehensive platform for debt tracking, budget recommendations, financial literacy resources, and goal-setting tools. It is structured into three primary layers: the Presentation Layer, Application Layer, and Data Layer.

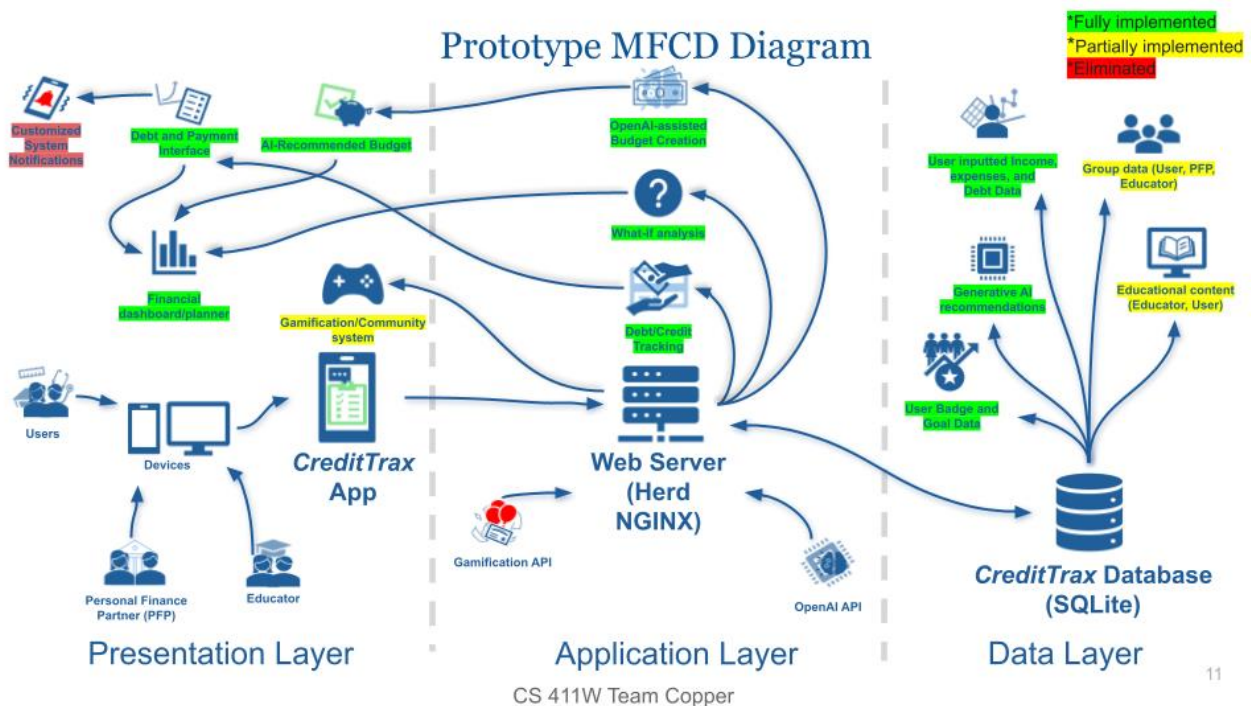
The Presentation Layer includes the user-facing components. Users interact with the system through a financial dashboard and planner, a debt and payment interface, and a gamification/community system. The Application Layer manages the core processing logic using a web server (Herd NGINX) to handle requests and process data. The system also incorporates a gamification framework to support achievement-based incentives and community engagement.

The Data Layer consists of the CreditTrax database, built using SQLite, which stores essential financial data such as user-inputted income, expenses, and debt details. Additionally,

the database supports educational content and user progress tracking through badges and goal data. The Major Functional Component Diagram (Figure 1) illustrates how these subsystems interact.

Figure 1

Major Functional Components Diagram



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Users input financial information through the CreditTrax application, which is processed by the web server and stored in the database. AI-driven analytics assist with budget creation, debt-tracking insights, and financial projections. The gamification system enhances user engagement by awarding badges for financial milestones.

2.2 Product Functions

The CreditTrax application encompasses several key financial management features, each designed to assist users in tracking debt, budgeting effectively, setting financial goals, and

analyzing future financial scenarios. Some features are fully implemented, while others remain partially developed or have been eliminated in the prototype. They are outlined in the RWP vs Prototype Table (Table 1) below.

Table 1

RWP vs Prototype Table

Real World Product vs Prototype

*Fully implemented
*Partially implemented
*Eliminated

Functionality Group	Feature	RWP	Prototype	Notes
Account Management	Add/Delete Account	Full add/delete functionality for user accounts	Fully Implemented	Can use stand-in accounts
	Login/Authentication	MFA and session management	Fully Implemented	Use stand-in accounts
	Profile Management	Customizable user profile	Fully Implemented	Use stand-in accounts
Debt Tracking	Add/Edit Debt	Comprehensive tracking editing	Fully Implemented	
	Smart Payment Reminders	Customizable with escalation	Eliminated	Unsure of method used to send reminders to user without Google APIs
Budgeting	Personalized Budget Creation	AI-driven recommendations	Fully Implemented	Focus generative AI use in what-if analysis
	Expense Tracking	Detailed tracking with visuals	Fully Implemented	
	Budget Adjustments	AI-driven recommendations based on spending patterns	Fully Implemented	Focus generative AI use in what-if analysis
Functionality Group	Feature	RWP	Prototype	Notes
Goal Tracking	Short/Long-Term Goals	Customizable short-term goal tracking	Fully Implemented	
	Goal Progress Visualization	Visual tracking	Fully Implemented	
What-If Analysis	AI-assisted projection/simulation	Use of AI for what-if	Fully Implemented	
	Create/Simulate scenarios	Algorithmic Impact simulations	Fully Implemented	
Analytics	Comprehensive Financial Dashboard	Budget and debt visualizations, goal progression,	Fully Implemented	
Reward System	Badges and Rewards	Custom badges for achievements	Partially Implemented	Basic badges for demonstration
	Community/Group page	Full community functionality with leaderboards	Partially Implemented	Can create groups
Financial Literacy Tools	Educational Resources	Upload external resources, show financial concepts embedded in other features	Partially Implemented	Only shows embedded finance information.

Debt Tracking allows users to comprehensively manage their debts by adding new obligations and editing existing ones. While the originally planned smart payment reminders with escalating notifications have been eliminated, a basic reminder system remains in place to assist users in staying on top of their payments.

Budgeting features include AI-assisted personalized budget recommendations based on user input, helping individuals optimize their financial planning. While the AI can't fully create a budget on its own, it provides valuable guidance to support users in developing their own. Additionally, expense tracking with visual representations is fully implemented, allowing users to monitor their spending habits effectively.

Goal Tracking is a core feature that is fully implemented, enabling users to set and manage both short-term and long-term financial goals. The system also includes goal and progress visualization, helping users track their achievements over time.

“What-If” Analysis provides AI-driven financial projections, which are fully implemented, allowing users to assess different financial scenarios. The feature includes both dynamic AI-generated projections and algorithm-based hardcoded scenarios to help users make informed financial decisions.

The Rewards System introduces gamification elements to encourage financial responsibility. Custom badges for achievements are partially implemented, with only basic badges available in the prototype. Additionally, the full community functionality, including leaderboards, remains partially implemented.

2.3 User Characteristics

CreditTrax serves two primary user roles: young adults and personal finance partners. In this prototype, we will primarily focus on the young adult user role, emphasizing features that

help them manage their finances by tracking debts and utilizing the “What-If” analysis tool. They can set financial goals, monitor progress through visualizations, and explore different financial scenarios to improve decision-making.

The functionalities for personal finance partners (such as financial advisors or wellness coaches) will not be implemented in this prototype. Personal finance partners typically assist users by tracking financial goals and modifying AI-generated budget plans, offering personalized guidance based on user spending trends to enhance financial stability.

The application must accommodate varying financial literacy levels with an intuitive interface. AI-assisted recommendations rely on accurate user data, and finance partners will have limited access to maintain privacy. External factors like economic shifts may also impact budgeting strategies.

2.4 Constraints

Legal constraints include framing financial insights as budget recommendations rather than budget advice to minimize liability.

Technical constraints involve the AI’s reliance on user input, which affects the accuracy of recommendations. Additionally, online and external data sources must be verified to ensure reliability and correctness.

2.5 Assumptions and Dependencies

The system depends on several third-party services and frameworks. OpenAI is used for AI-driven budgeting recommendations. The frameworks supporting development include Laravel (backend for PHP), Tailwind CSS (styling), and Alpine.js (frontend interactions). Laravel-gamify framework is also used for gamification features. Additionally, the project relies

on Composer as a dependency manager and Chart.js for data visualizations.